

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
11	(a)	(i)		large increase after removing the 2 nd electron			1	1		
		(ii)		student is partially correct / incorrect (can be inferred from answer) – must have some sensible attempt at explanation (1) X is higher than Y since X has a larger nuclear charge but little/no extra shielding (1) X is not lower than Z since the outer electron in Z is partly shielded by the s electrons (and therefore more easily lost) (1) reference to stronger attraction is neutral			3	3		
	(b)	(i)		Na ⁺ (g) → Na ²⁺ (g) + e ⁻	1			1		
		(ii)		$\Delta E = \frac{4560000}{6.02 \times 10^{23}} = 7.57 \times 10^{-18} \text{ J} \quad (1)$ $f = \frac{\Delta E}{h} = \frac{7.57 \times 10^{-18}}{6.63 \times 10^{-34}} = 1.14 \times 10^{16} \text{ s}^{-1} \quad (1)$ $\lambda = \frac{3.00 \times 10^8}{1.14 \times 10^{16}} = 2.63 \times 10^{-8} \text{ m} \quad (1)$ $\lambda = 26.3 \text{ nm} \quad (1)$ ecf possible throughout	1	3		4	4	